Teacher Wu Kuan | Exam at a Glance #26 — Standard Deviation Under a Linear Change

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1. The trap question

Everyone gets +10 points. What happens to the standard deviation?

2. The method

- SD measures spread. Adding a constant shifts the data: mean + b, SD unchanged.
- Multiplying by a scales every distance: SD × |a|, variance × a^2.
- For y = ax + b: mean(y) = a·mean(x) + b, SD(y) = |a|·SD(x).

3. Parent check method

Ask: “If the whole class moves 10 steps to the right, are they any more spread out?”

Practice

1. Data: 2, 4, 4, 4, 5, 5, 7, 9. Find the mean and standard deviation.
2. Add 10 to every value in Problem 1. New mean and SD?
3. Data x has mean 5 and SD 2. Let y = 3x − 5. Find the mean and SD of y.
4. Data x has mean 5 and SD 2. Let y = −2x + 1. Find the SD and variance of y.
5. A test has mean 40 and SD 10. The teacher rescales with y = ax + b (a > 0) to get mean 60 and SD 15. Find a and b.

Answers

1. Mean 5, SD 2 (squared deviations sum to 32; 32 ÷ 8 = 4)
2. Mean 15, SD 2
3. Mean 10, SD 6
4. SD 4, variance 16
5. a = 3/2, b = 0

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